Supplementary Material

A Multimodal Framework for Subtype Prediction Using Histopathology and Transcriptomics in Breast Cancer

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This document contains 15 supplementary tables and 12 supplementary figures referenced in the main text. The source code, the GDC download commands and the preprocessing scripts for the five approaches are available at https://github.com/dlagunasantof/tesis-brca-multimodal.

Supplementary Tables

Table S1. Classification on the full filtered transcriptome (17,324 genes).

Feature selection was applied inside each model pipeline with SelectKBest and the ANOVA F statistic (f_classif), so that genes were selected using only the training data of each partition. The number of retained genes (k) was optimized as a hyperparameter: {200, 500, 1,000} for KNN and {100, 200, 400, all} for XGBoost, yielding k = 1,000 and k = all respectively. Random Forest, Logistic Regression and SVM were trained on the full filtered transcriptome.

Classifier	Accuracy	Precision (macro)	Sensitivity (macro)	Macro-F1
Logistic Regression	0.946	0.95	0.95	0.95
XGBoost	0.934	0.94	0.94	0.94
Random Forest	0.928	0.93	0.93	0.93
SVM	0.928	0.93	0.93	0.93
KNN	0.873	0.88	0.88	0.88

Table S1a. Performance on the hold-out test set (n = 166).

Classifier	Macro-F1	Balanced accuracy	Precision (macro)	Sensitivity (macro)
XGBoost	0.934 ± 0.017	0.935 ± 0.017	0.936 ± 0.017	0.935 ± 0.017
Logistic Regression	0.929 ± 0.019	0.928 ± 0.019	0.930 ± 0.018	0.928 ± 0.019
Random Forest	0.923 ± 0.019	0.923 ± 0.020	0.926 ± 0.018	0.923 ± 0.020
SVM	0.914 ± 0.024	0.914 ± 0.024	0.916 ± 0.024	0.914 ± 0.024
KNN	0.879 ± 0.022	0.880 ± 0.022	0.888 ± 0.018	0.880 ± 0.022

Table S1b. Performance under repeated stratified 5-fold cross-validation with 3 repetitions (n = 552; mean ± SD). The Wilcoxon signed-rank test between the two best classifiers (XGBoost vs Logistic Regression) gave p = 0.347. Compared with the 604-gene consensus signature (macro-F1 = 0.928-0.934 in cross-validation), the full transcriptome reached a similar performance, indicating that the 604-gene signature retains virtually all the discriminative information of the transcriptome.

Table S2. Transcriptomic preprocessing and quality control.

Step	Genes	Samples
Raw matrix (GDC, FPKM_unstranded, GENCODE v36, protein-coding)	19,964	552
log2(FPKM + 1) transformation	19,964	552
Removal of genes with zero total expression, > 50% zero values or zero variance	17,324	552
Outlier inspection of sample distributions	17,324	552
QC-filtered transcriptome (input to limma and to the classifiers)	17,324	552

Step	Genes	Samples
TOP1000 most variable genes (input to the consensus signature)	1,000	552
Consensus DEG signature (union of both schemes, unique genes)	604	552

Table S2. Number of genes and samples retained at each preprocessing step of the transcriptomic branch (TCGA-BRCA).

Table S3. Hyperparameter search spaces.

The search spaces were not identical across representations. Four configurations were used, listed below. In every case the search was performed exclusively on the training set (n = 386) under stratified 5-fold cross-validation (random_state = 42) with refit on the best configuration.

Configuration	Representations	Objective	Randomized-search iterations
A	Consensus signature (604 genes); full filtered transcriptome (17,324 genes)	macro-F1	RF 50; XGBoost 60
B	E1 KimiaNet-Mosaic (4 distance metrics)	macro-F1	RF 50; XGBoost 50
C	E2, E3, E4, E5	accuracy	RF 50; XGBoost 60
D	MOFA+ (RNA + WSI); MOFA RNA-only; MOFA WSI-only	macro-F1	RF 50; XGBoost 50

Table S3a. The four search configurations and the representations that used each one.

Classifier	Hyperparameter	A (transcriptomic)	B (E1)	C (E2-E5)	D (MOFA)
KNN	n_neighbors	1-30	1-30	1-30	1-30
	weights	uniform, distance	uniform, distance	uniform, distance	uniform, distance
	metric	minkowski, cosine, euclidean	minkowski, cosine	minkowski, cosine	minkowski, cosine
	p	1, 2, 3	1, 2, 3	1, 2, 3	1, 2, 3
	StandardScaler	yes	yes	no	yes
Random Forest	n_estimators	100, 200, 300, 500	idem A	idem A	idem A
	max_depth	None, 30, 40, 50	idem A	idem A	idem A
	min_samples_split	2, 5, 10	idem A	idem A	idem A
	min_samples_leaf	1, 2, 5	idem A	idem A	idem A
	max_features	0.02, 0.05, 0.10, 0.15, 0.18	idem A	idem A	idem A
	criterion	gini, entropy	idem A	idem A	idem A
Logistic Regression	C	0.001, 0.01, 0.1, 1	0.01, 0.1, 1, 10	0.001, 0.01, 0.1, 1	0.01, 0.1, 1, 10
	penalty	l1, l2	l2	l1, l2	l2
	solver	liblinear, lbfgs, newton-cg, saga	lbfgs	liblinear, lbfgs, newton-cg, saga	lbfgs
	multi_class	ovr, multinomial	not tuned	ovr, multinomial	not tuned
SVM	C	0.01, 0.1, 1, 10, 100	idem A	idem A	idem A
	kernel	poly, rbf	idem A	idem A	idem A
	gamma	scale, 1, 0.1, 0.01, 0.001, 0.0001	idem A	idem A	idem A
	degree	2, 3, 4	idem A	idem A	idem A
	coef0	0.0, 0.1, 1.0	idem A	idem A	idem A
XGBoost	sel__k (SelectKBest)	100, 200, 400, all	not used	100, 200, 400, all	not used
	n_estimators	U{200, 599}	U{200, 599}	U{200, 599}	U{200, 599}
	max_depth	U{3, 7}	U{3, 6}	U{3, 7}	U{3, 6}
	learning_rate	U(0.05, 0.35)	U(0.03, 0.18)	U(0.05, 0.35)	U(0.03, 0.18)
	subsample	U(0.6, 0.9)	U(0.6, 0.9)	U(0.6, 0.9)	U(0.6, 0.9)
	colsample_bytree	U(0.4, 0.8)	U(0.4, 0.9)	U(0.4, 0.8)	U(0.4, 0.9)
	min_child_weight	U{1, 4}	U{1, 5}	U{1, 4}	U{1, 5}
	reg_lambda	U(0.5, 2.5)	not used	U(0.5, 2.5)	not used
	reg_alpha	U(0.0, 1.0)	not used	U(0.0, 1.0)	not used

Table S3b. Hyperparameter values explored in each configuration. U{a, b} denotes a uniform integer distribution over [a, b] and U(a, b) a continuous uniform distribution over [a, b]; "idem A" means identical to configuration A. Three caveats follow from this table. First, configuration C optimized accuracy rather than macro-F1; with three near-balanced classes (191/195/166) both criteria select very similar configurations, and the macro-F1 values reported throughout the study are computed on the predictions of the selected model, not on the search objective. Second, KNN was fitted without standardization in configuration C. Third, in configurations B and D the penalty and solver of Logistic Regression had a single option, so only C was effectively tuned.

Table S4. Selected hyperparameter values.

Representation	Conf.	Classifier	Selected values
Full filtered transcriptome (17,324 genes)	A	KNN	sel__k = 1,000; metric = euclidean; n_neighbors = 3; weights = uniform
		Random Forest	criterion = entropy; max_depth = 40; max_features = 0.15; min_samples_leaf = 5; min_samples_split = 10; n_estimators = 500
		Logistic Regression	C = 1; multi_class = multinomial; penalty = l1; solver = saga
		SVM	C = 1; kernel = poly; degree = 2; gamma = scale; coef0 = 1.0
		XGBoost	sel__k = all; colsample_bytree = 0.759; learning_rate = 0.348; max_depth = 6; min_child_weight = 1; n_estimators = 267; reg_alpha = 0.237; reg_lambda = 1.151; subsample = 0.824
Consensus signature (604 genes)	A	KNN	metric = minkowski; p = 1; n_neighbors = 17; weights = uniform
		Random Forest	criterion = entropy; max_depth = 50; max_features = 0.15; min_samples_leaf = 2; min_samples_split = 10; n_estimators = 300
		Logistic Regression	C = 1; multi_class = ovr; penalty = l1; solver = liblinear
		SVM	C = 1; kernel = poly; degree = 2; gamma = scale; coef0 = 1.0
		XGBoost	sel__k = 200; colsample_bytree = 0.557; learning_rate = 0.071; max_depth = 4; min_child_weight = 1; n_estimators = 441; reg_alpha = 0.683; reg_lambda = 0.642; subsample = 0.696
E1 KimiaNet-Mosaic (Euclidean)	B	KNN	metric = cosine; n_neighbors = 24; weights = distance
		Random Forest	criterion = entropy; max_depth = 50; max_features = 0.18; min_samples_leaf = 2; min_samples_split = 5; n_estimators = 300
		Logistic Regression	C = 0.01; penalty = l2; solver = lbfgs
		SVM	C = 10; kernel = poly; degree = 2; gamma = 0.0001; coef0 = 1.0
		XGBoost	colsample_bytree = 0.725; learning_rate = 0.038; max_depth = 6; min_child_weight = 4; n_estimators = 493; subsample = 0.600
E2 KimiaNet-Crop	C	KNN	metric = cosine; n_neighbors = 16; weights = uniform
		Random Forest	criterion = gini; max_depth = 50; max_features = 0.18; min_samples_leaf = 1; min_samples_split = 10; n_estimators = 500
		Logistic Regression	C = 0.001; multi_class = multinomial; penalty = l2; solver = lbfgs
		SVM	C = 1; kernel = poly; degree = 3; gamma = 0.0001; coef0 = 1.0
		XGBoost	sel__k = 100; colsample_bytree = 0.495; learning_rate = 0.268; max_depth = 7; min_child_weight = 4; n_estimators = 297; reg_alpha = 0.634; reg_lambda = 1.572; subsample = 0.627
E3 UNI-Crop	C	KNN	metric = minkowski; p = 1; n_neighbors = 30; weights = distance
		Random Forest	criterion = gini; max_depth = None; max_features = 0.05; min_samples_leaf = 2; min_samples_split = 2; n_estimators = 500
		Logistic Regression	C = 0.001; multi_class = multinomial; penalty = l2; solver = saga
		SVM	C = 10; kernel = rbf; gamma = 0.0001
		XGBoost	sel__k = 100; colsample_bytree = 0.780; learning_rate = 0.222; max_depth = 7; min_child_weight = 1; n_estimators = 298; reg_alpha = 0.602; reg_lambda = 1.607; subsample = 0.664
E4 KimiaNet-KMedoids	C	KNN	metric = minkowski; p = 1; n_neighbors = 9; weights = distance
		Random Forest	criterion = gini; max_depth = 50; max_features = 0.10; min_samples_leaf = 1; min_samples_split = 10; n_estimators = 100
		Logistic Regression	C = 0.01; multi_class = ovr; penalty = l2; solver = lbfgs
		SVM	C = 0.01; kernel = poly; degree = 2; gamma = 0.01; coef0 = 1.0
		XGBoost	sel__k = 400; colsample_bytree = 0.504; learning_rate = 0.349; max_depth = 6; min_child_weight = 1; n_estimators = 236; reg_alpha = 0.883; reg_lambda = 0.877; subsample = 0.684

Representation	Conf.	Classifier	Selected values
E5 UNI2-h-KMedoids	C	KNN	metric = minkowski; p = 1; n_neighbors = 16; weights = uniform
		Random Forest	criterion = entropy; max_depth = 50; max_features = 0.05; min_samples_leaf = 1; min_samples_split = 10; n_estimators = 200
		Logistic Regression	C = 1; multi_class = ovr; penalty = l2; solver = liblinear
		SVM	C = 10; kernel = poly; degree = 2; gamma = 0.0001; coef0 = 1.0
		XGBoost	sel__k = 200; colsample_bytree = 0.468; learning_rate = 0.070; max_depth = 6; min_child_weight = 1; n_estimators = 515; reg_alpha = 0.563; reg_lambda = 1.271; subsample = 0.605
MOFA+ (RNA + WSI)	D	KNN	metric = minkowski; p = 1; n_neighbors = 28; weights = uniform
		Random Forest	criterion = gini; max_depth = 30; max_features = 0.18; min_samples_leaf = 1; min_samples_split = 2; n_estimators = 100
		Logistic Regression	C = 1; penalty = l2; solver = lbfgs
		SVM	C = 1; kernel = rbf; gamma = 0.01
		XGBoost	colsample_bytree = 0.612; learning_rate = 0.089; max_depth = 6; min_child_weight = 3; n_estimators = 280; subsample = 0.813
MOFA RNA-only	D	KNN	metric = minkowski; p = 3; n_neighbors = 18; weights = distance
		Random Forest	criterion = gini; max_depth = 30; max_features = 0.18; min_samples_leaf = 1; min_samples_split = 2; n_estimators = 100
		Logistic Regression	C = 0.1; penalty = l2; solver = lbfgs
		SVM	C = 10; kernel = poly; degree = 3; gamma = 0.01; coef0 = 1.0
		XGBoost	colsample_bytree = 0.725; learning_rate = 0.038; max_depth = 6; min_child_weight = 4; n_estimators = 493; subsample = 0.600
MOFA WSI-only	D	KNN	metric = minkowski; p = 3; n_neighbors = 10; weights = uniform
		Random Forest	criterion = entropy; max_depth = 30; max_features = 0.05; min_samples_leaf = 5; min_samples_split = 5; n_estimators = 100
		Logistic Regression	C = 10; penalty = l2; solver = lbfgs
		SVM	C = 100; kernel = poly; degree = 3; gamma = 1; coef0 = 0.1
		XGBoost	colsample_bytree = 0.739; learning_rate = 0.032; max_depth = 5; min_child_weight = 3; n_estimators = 347; subsample = 0.794

Table S4. Hyperparameter values selected by the search described in Table S3, for the ten representations evaluated. The "Conf." column refers to the search configuration of Table S3a. Values in brackets were omitted when the selected kernel or metric makes them inoperative (p under a cosine metric, degree and coef0 under an RBF kernel). The CV score is the cross-validated macro-F1 of the winning configuration within the search (search.best_score_, 5 folds over the 386 training samples); it is an optimistically biased estimate because it is a maximum over many configurations, and it was only recorded for the MOFA representations. XGBoost for E1 and for MOFA RNA-only converged to the same configuration because both drew from the same distributions with the same seed and iteration count.

Table S5. Multilayer perceptron: architecture, training configuration and results.

Component	Value
Input representation	E3 UNI-Crop embeddings (1,024 dimensions)
Architecture	1,024 → 512 → 256 → 128 → 3
Hidden layers	Batch normalization, ReLU and dropout (0.4)
Loss function	Cross-entropy weighted inversely to class frequency
Optimizer	AdamW (learning rate 1 × 10-3; weight decay 1 × 10-4)
Learning-rate schedule	Cosine annealing
Epochs / batch size	150 / 32
Standardization	Fitted on the training set only
Framework	PyTorch 2.5.1 (CUDA 12.1)
Macro-F1 (test)	0.692

Table S5. Multilayer perceptron trained on the best histopathological representation (E3). The epoch was selected on the test set, which yields an optimistic estimate of its performance (see Limitations).

Table S6. Differentially expressed genes, one-vs-rest scheme.

Contrast	Filtered transcriptome FDR < 0.05	Filtered transcriptome + |log2FC| > 1	TOP1000 FDR < 0.05	TOP1000 + |log2FC| > 1
Luminal A vs rest	11,516	391	819	285
Luminal B vs rest	10,978	253	778	229

Contrast	Filtered transcriptome FDR < 0.05	Filtered transcriptome + |log2FC| > 1	TOP1000 FDR < 0.05	TOP1000 + |log2FC| > 1
Basal-like vs rest	13,479	1,107	912	577

Table S6. Number of differentially expressed genes per subtype in the one-vs-rest scheme (limma), in the QC-filtered transcriptome (17,324 genes) and in the 1,000 most variable genes.

Table S7. Differentially expressed genes, one-vs-one scheme.

Contrast	Filtered transcriptome FDR < 0.05	Filtered transcriptome + |log2FC| > 1	TOP1000 FDR < 0.05	TOP1000 + |log2FC| > 1
Luminal A vs Luminal B	8,025	130	601	76
Luminal A vs Basal-like	13,086	1,284	898	606
Luminal B vs Basal-like	12,812	1,076	884	553

Table S7. Number of differentially expressed genes in the one-vs-one scheme (limma). The Luminal A versus Luminal B contrast yields the fewest genes in every configuration.

Table S8. Per-subtype metrics, transcriptomic branch, test set.

Classifier	LumA Prec.	LumA Sens.	LumA F1	LumB Prec.	LumB Sens.	LumB F1	Basal Prec.	Basal Sens.	Basal F1
Logistic Regression	0.93	0.89	0.91	0.90	0.93	0.92	0.98	0.98	0.98
SVM (polynomial kernel)	0.88	0.91	0.90	0.91	0.88	0.90	0.98	0.98	0.98
XGBoost	0.86	0.86	0.86	0.88	0.86	0.87	0.98	1.00	0.99
Random Forest	0.86	0.84	0.85	0.86	0.86	0.86	0.98	1.00	0.99
KNN	0.82	0.82	0.82	0.82	0.83	0.82	1.00	0.98	0.99

Table S8. Precision, sensitivity and F1 per subtype on the hold-out test set using the 604-gene consensus signature (LumA n = 57; LumB n = 59; Basal-like n = 50).

Table S9. Genes per subtype and direction of expression in each scheme.

Subtype and direction	One-vs-rest	One-vs-one	Intersection
UP Luminal A	160	30	29
DOWN Luminal A	32	9	9
UP Luminal B	40	0	0
DOWN Luminal B	97	28	25
UP Basal-like	236	199	199
DOWN Basal-like	340	289	289
Total	905	555	551

Table S9. Genes per subtype and direction of expression in each differential-expression scheme and in their intersection (TOP1000). The consensus signature is the union of both schemes (905 + 555 - 551 = 909 entries, 604 unique genes); the intersection of 551 entries is reported only as a consistency measure. No commonly upregulated gene was found for Luminal B across both schemes.

Table S10. Transcriptomic branch, repeated cross-validation.

Classifier	Macro-F1	Balanced accuracy	Precision (macro)	Sensitivity (macro)
XGBoost	0.934 ± 0.020	0.934 ± 0.020	0.935 ± 0.020	0.934 ± 0.020
Logistic Regression	0.928 ± 0.022	0.928 ± 0.022	0.930 ± 0.022	0.928 ± 0.022
Random Forest	0.919 ± 0.025	0.919 ± 0.025	0.922 ± 0.025	0.919 ± 0.025
SVM	0.917 ± 0.025	0.917 ± 0.025	0.919 ± 0.025	0.917 ± 0.025
KNN	0.900 ± 0.034	0.900 ± 0.034	0.901 ± 0.033	0.900 ± 0.034

Table S10. Performance of the classifiers on the 604-gene consensus signature under repeated stratified 5-fold cross-validation with 3 repetitions (n = 552; mean ± SD). The difference between XGBoost and Logistic Regression was not significant (Wilcoxon signed-rank test on per-fold macro-F1, p = 0.252).

Table S11. Histopathological branch: detailed results per approach.

E1 KimiaNet-Mosaic. Macro-F1 by classifier and distance metric used in the K-Medoids patch selection.

Classifier	Cosine	Euclidean	Bhattacharyya	Jensen-Shannon	Mean
Logistic Regression	0.649	0.652	0.629	0.636	0.642
SVM	0.613	0.622	0.589	0.614	0.609

Classifier	Cosine	Euclidean	Bhattacharyya	Jensen-Shannon	Mean
XGBoost	0.592	0.616	0.580	0.594	0.595
Random Forest	0.592	0.615	0.582	0.585	0.593
KNN	0.488	0.513	0.487	0.525	0.503

Table S11a. E1: macro-F1 on the test set (n = 166) by classifier and distance metric.

Classifier	Cosine	Euclidean	Bhattacharyya	Jensen-Shannon
SVM	0.654 ± 0.048	0.659 ± 0.051	0.642 ± 0.042	0.653 ± 0.040
Logistic Regression	0.649 ± 0.047	0.655 ± 0.043	0.642 ± 0.042	0.652 ± 0.043
XGBoost	0.638 ± 0.054	0.636 ± 0.039	0.633 ± 0.047	0.654 ± 0.050
Random Forest	0.599 ± 0.044	0.604 ± 0.045	0.608 ± 0.039	0.599 ± 0.042
KNN	0.544 ± 0.045	0.558 ± 0.043	0.550 ± 0.032	0.557 ± 0.031

Table S11b. E1: macro-F1 under repeated cross-validation (mean ± SD). The Euclidean metric was the best overall and was the one carried forward.

E2 KimiaNet-Crop.

Classifier	Accuracy (test)	Macro-F1 (test)	Macro-F1 (CV)	Balanced acc. (CV)	Precision (CV)	Sensitivity (CV)
Logistic Regression	0.452	0.452	0.460 ± 0.033	0.462 ± 0.031	0.471 ± 0.035	0.462 ± 0.031
Random Forest	0.446	0.448	0.445 ± 0.038	0.448 ± 0.036	0.450 ± 0.040	0.448 ± 0.036
XGBoost	0.386	0.390	0.441 ± 0.038	0.444 ± 0.035	0.443 ± 0.039	0.444 ± 0.035
SVM	0.452	0.442	0.433 ± 0.031	0.438 ± 0.029	0.450 ± 0.036	0.438 ± 0.029
KNN	0.428	0.411	0.427 ± 0.046	0.431 ± 0.044	0.435 ± 0.046	0.431 ± 0.044

Table S11c. E2 KimiaNet-Crop.

E3 UNI-Crop.

Classifier	Accuracy (test)	Macro-F1 (test)	Macro-F1 (CV)	Balanced acc. (CV)	Precision (CV)	Sensitivity (CV)
Logistic Regression	0.705	0.708	0.741 ± 0.034	0.744 ± 0.034	0.743 ± 0.035	0.744 ± 0.034
SVM	0.705	0.707	0.734 ± 0.041	0.736 ± 0.042	0.736 ± 0.041	0.736 ± 0.042
XGBoost	0.651	0.654	0.729 ± 0.041	0.730 ± 0.040	0.731 ± 0.042	0.730 ± 0.040
Random Forest	0.663	0.667	0.719 ± 0.043	0.718 ± 0.044	0.726 ± 0.045	0.718 ± 0.044
KNN	0.651	0.651	0.643 ± 0.047	0.644 ± 0.046	0.646 ± 0.047	0.644 ± 0.046

Table S11d. E3 UNI-Crop, the best-performing histopathological representation.

E4 KimiaNet-KMedoids.

Classifier	Accuracy (test)	Macro-F1 (test)	Macro-F1 (CV)	Balanced acc. (CV)	Precision (CV)	Sensitivity (CV)
Logistic Regression	0.494	0.496	0.451 ± 0.049	0.453 ± 0.048	0.459 ± 0.051	0.453 ± 0.048
XGBoost	0.392	0.392	0.438 ± 0.039	0.440 ± 0.039	0.440 ± 0.040	0.440 ± 0.039
SVM	0.458	0.443	0.428 ± 0.037	0.434 ± 0.036	0.463 ± 0.049	0.434 ± 0.036
Random Forest	0.398	0.399	0.416 ± 0.039	0.420 ± 0.039	0.420 ± 0.041	0.420 ± 0.039
KNN	0.386	0.385	0.391 ± 0.034	0.390 ± 0.035	0.397 ± 0.034	0.390 ± 0.035

Table S11e. E4 KimiaNet-KMedoids.

E5 UNI2-h-KMedoids.

Classifier	Accuracy (test)	Macro-F1 (test)	Macro-F1 (CV)	Balanced acc. (CV)	Precision (CV)	Sensitivity (CV)
SVM	0.687	0.688	0.733 ± 0.041	0.732 ± 0.043	0.739 ± 0.040	0.732 ± 0.043
Logistic Regression	0.669	0.671	0.730 ± 0.030	0.733 ± 0.028	0.735 ± 0.031	0.733 ± 0.028
XGBoost	0.633	0.636	0.691 ± 0.037	0.693 ± 0.039	0.693 ± 0.036	0.693 ± 0.039
Random Forest	0.663	0.665	0.678 ± 0.040	0.678 ± 0.042	0.682 ± 0.039	0.678 ± 0.042
KNN	0.548	0.556	0.559 ± 0.041	0.553 ± 0.044	0.582 ± 0.033	0.553 ± 0.044

Table S11f. E5 UNI2-h-KMedoids. Repeated cross-validation values are the mean ± SD over 5 folds with 3 repetitions (n = 552).

Macro-averaged precision and sensitivity on the test set, complementing the tables above.

Approach	Classifier	Accuracy	Precision (macro)	Sensitivity (macro)	Macro-F1
E2 KimiaNet-Crop	Logistic Regression	0.452	0.464	0.449	0.452
	Random Forest	0.446	0.449	0.447	0.448
	SVM	0.452	0.454	0.446	0.442
	KNN	0.428	0.418	0.420	0.411
	XGBoost	0.386	0.394	0.388	0.390
E3 UNI-Crop	Logistic Regression	0.705	0.705	0.713	0.708
	SVM	0.705	0.704	0.714	0.707
	Random Forest	0.663	0.668	0.668	0.667
	XGBoost	0.651	0.652	0.658	0.654
	KNN	0.651	0.648	0.661	0.651
E4 KimiaNet-KMedoids	Logistic Regression	0.494	0.498	0.496	0.496
	SVM	0.458	0.469	0.451	0.443
	Random Forest	0.398	0.398	0.399	0.399
	XGBoost	0.392	0.392	0.395	0.392
	KNN	0.386	0.398	0.384	0.385
E5 UNI2-h-KMedoids	SVM	0.687	0.687	0.696	0.688
	Logistic Regression	0.669	0.670	0.676	0.671
	Random Forest	0.663	0.664	0.671	0.665
	XGBoost	0.633	0.633	0.642	0.636
	KNN	0.548	0.563	0.553	0.556

Table S11g. Accuracy, macro-averaged precision and sensitivity, and macro-F1 of each classifier on the hold-out test set (n = 166) for approaches E2 to E5.

Table S12. Patient-to-patient similarity matrices.

Property	Transcriptomics (S_RNA)	Histopathology (S_WSI)
Dimension	552 × 552	552 × 552
Metric	Pearson correlation	Cosine similarity on centered embeddings
Value range	[-0.641; 1.000]	[-0.440; 1.000]
Global gap (mean of the three subtype gaps)	0.4364	0.0477
Basal-like gap	0.7938 ($p < 0.001$)	0.0755 ($p < 0.001$)
Luminal A gap	0.2654 ($p < 0.001$)	0.0494 ($p < 0.001$)
Luminal B gap	0.2501 ($p < 0.001$)	0.0183 ($p = 4.56 \times 10-26$)
Subtype structure	Strong	Weak

Table S12. Comparison of the patient-to-patient similarity matrices between modalities. The gap statistic is the mean intra-subtype similarity minus the mean inter-subtype similarity, computed on S in [-1, 1]; significance was assessed with the Mann-Whitney U test on unique non-redundant pairs. The histopathological matrix is built on the E3 (UNI-Crop) embeddings, L2-normalized and then centered per feature, the same preprocessing used for the WSI view in MOFA+. Cosine similarity does not centre its inputs, whereas Pearson correlation does; computing it on uncentered embeddings leaves the component shared by all patients in every pair and compresses the subtype structure. On uncentered embeddings the same statistics gave a value range of [-0.068; 1.000] and a global gap of 0.0287.

Table S13. Mantel test.

Set	n	Pairs	r	r2	p-value
Global	552	152,076	0.174	3.0%	≤ 0.0001
Basal-like	166	13,695	0.100	1.0%	≤ 0.0001
Luminal A	191	18,145	0.131	1.7%	≤ 0.0001
Luminal B	195	18,915	0.062	0.4%	≤ 0.0001

Table S13. Mantel test between the transcriptomic (consensus signature) and histopathological (UNI-Crop) distance matrices, globally and per subtype (9,999 permutations). The coupling is significant in all three subtypes but uniformly small. Computed on uncentered

embeddings, the same test gave a global r of 0.112 and a different per-subtype ordering, with Luminal B not significant; that pattern was an artefact of the component shared by all embeddings.

Table S14. MOFA+ factors.

Factor	R2 RNA (%)	R2 WSI (%)	Type	F	FDR	FDR < 0.05
F1	32.90	2.29	Shared	3,408	4.23 × 10-309	Yes
F2	0.02	7.82	WSI-specific	4.188	0.0274	Yes
F3	0.02	6.75	WSI-specific	0.753	0.4715	No
F4	5.20	0.04	RNA-specific	98.41	2.09 × 10-36	Yes
F5	0.04	4.53	WSI-specific	2.551	0.1005	No
F6	0.06	4.02	WSI-specific	1.426	0.2597	No
F7	1.06	2.78	Shared	33.24	8.26 × 10-14	Yes
F8	0.95	2.75	Uncategorized	22.26	1.42 × 10-9	Yes
F9	2.19	1.50	Shared	38.14	1.45 × 10-15	Yes
F10	0.01	3.49	WSI-specific	2.648	0.1005	No
F11	0.00	3.48	WSI-specific	1.829	0.1886	No
F12	0.88	2.30	Uncategorized	10.96	4.30 × 10-5	Yes
F13	0.00	3.16	WSI-specific	13.63	3.89 × 10-6	Yes
F14	0.04	2.68	WSI-specific	2.573	0.1005	No
Total	43.29	46.34	3 shared, 1 RNA, 8 WSI, 2 uncat.			8 of 14

Table S14. Variance explained by each MOFA+ factor in each view, factor category and association with PAM50 subtype (one-way ANOVA with Benjamini-Hochberg FDR correction). A factor is shared when R2 > 1% in both views, RNA-specific when R2 > 1% in RNA and < 0.5% in WSI, and WSI-specific in the reverse case.

Table S15. MOFA+ single-view models.

Model	Factors	Total R2 (%)	Best classifier	Macro-F1 (test)
MOFA+ RNA + WSI	14	43.29 (RNA) / 46.34 (WSI)	Logistic Regression	0.907
MOFA RNA-only	14	58.44	Random Forest	0.896
MOFA WSI-only	10	0.00	Logistic Regression	0.446

Table S15a. Single-view MOFA+ models trained with an equivalent configuration to the multimodal model, reported as a complementary interpretability analysis.

Factor	MOFA RNA-only F	FDR	FDR < 0.05	MOFA WSI-only F	FDR	FDR < 0.05
F1	3,528.764	5.93 × 10-313	Yes	13.741	< 0.001	Yes
F2	171.247	1.12 × 10-57	Yes	13.746	< 0.001	Yes
F3	1.491	0.395	No	13.747	< 0.001	Yes
F4	0.045	0.965	No	13.865	< 0.001	Yes
F5	12.937	1.13 × 10-5	Yes	13.783	< 0.001	Yes
F6	7.308	2.07 × 10-3	Yes	13.740	< 0.001	Yes
F7	0.035	0.965	No	13.572	< 0.001	Yes
F8	0.928	0.554	No	13.416	< 0.001	Yes
F9	4.369	0.031	Yes	11.725	< 0.001	Yes
F10	0.533	0.747	No	13.735	< 0.001	Yes
F11	16.145	7.17 × 10-7	Yes	—	—	—
F12	3.807	0.046	Yes	—	—	—
F13	0.037	0.965	No	—	—	—
F14	1.193	0.473	No	—	—	—

Table S15b. Association of each single-view factor with PAM50 subtype (one-way ANOVA with Benjamini-Hochberg FDR correction). The RNA-only model retained 14 effective factors explaining 58.44% of the transcriptomic variance, 7 of them significantly associated with subtype and led by F1, which replicates the dominant factor of the multimodal model. The WSI-only model retained 10 factors but explained no variance (0.00%) despite the relaxed pruning configuration; all ten showed nearly identical F statistics (11.73-13.87), a pattern consistent with redundant factors reproducing the same residual signal rather than with informative histopathological structure.

MOFA model	Classifier	Macro-F1 (test)	Accuracy (test)	F1 Basal	F1 LumA	F1 LumB
MOFA RNA-only	Random Forest	0.896	0.892	0.99	0.839	0.857
MOFA RNA-only	Logistic Regression	0.895	0.892	0.98	0.847	0.860

MOFA model	Classifier	Macro-F1 (test)	Accuracy (test)	F1 Basal	F1 LumA	F1 LumB
MOFA RNA-only	XGBoost	0.890	0.886	0.99	0.835	0.845
MOFA RNA-only	SVM	0.884	0.880	0.98	0.832	0.840
MOFA RNA-only	KNN	0.832	0.825	0.97	0.761	0.767
MOFA WSI-only	Logistic Regression	0.446	0.452	0.482	0.504	0.351
MOFA WSI-only	SVM	0.417	0.446	0.438	0.538	0.275
MOFA WSI-only	XGBoost	0.411	0.416	0.358	0.472	0.404
MOFA WSI-only	Random Forest	0.392	0.398	0.372	0.435	0.370
MOFA WSI-only	KNN	0.369	0.373	0.327	0.425	0.355

Table S15c. Classification on the single-view latent spaces, hold-out test set (n = 166).

Classifier	MOFA RNA-only	MOFA+ (RNA + WSI)	Difference	MOFA WSI-only
SVM	0.917 ± 0.025	0.878 ± 0.027	+0.039	0.436 ± 0.036
Logistic Regression	0.914 ± 0.022	0.883 ± 0.026	+0.031	0.430 ± 0.036
Random Forest	0.908 ± 0.022	0.881 ± 0.026	+0.027	0.411 ± 0.030
XGBoost	0.905 ± 0.026	0.867 ± 0.019	+0.038	0.420 ± 0.043
KNN	0.876 ± 0.028	0.830 ± 0.033	+0.046	0.430 ± 0.041

Table S15d. Macro-F1 under repeated stratified 5-fold cross-validation with 3 repetitions (n = 552; mean ± SD) on the three MOFA latent spaces. The RNA-only space outperformed the multimodal one for all five classifiers, by 0.027 to 0.046. On the single hold-out partition the ordering was reversed (0.907 for the multimodal model versus 0.896 for the RNA-only model), but that comparison rests on 166 samples, whereas the cross-validated difference is consistent across every classifier. The WSI-only space obtained the lowest performance of all evaluated configurations. Both MOFA models were fitted on the complete cohort before partitioning, so all three sets of values are inflated to a similar extent; the comparison between them remains informative, their absolute magnitude does not.

Supplementary Figures

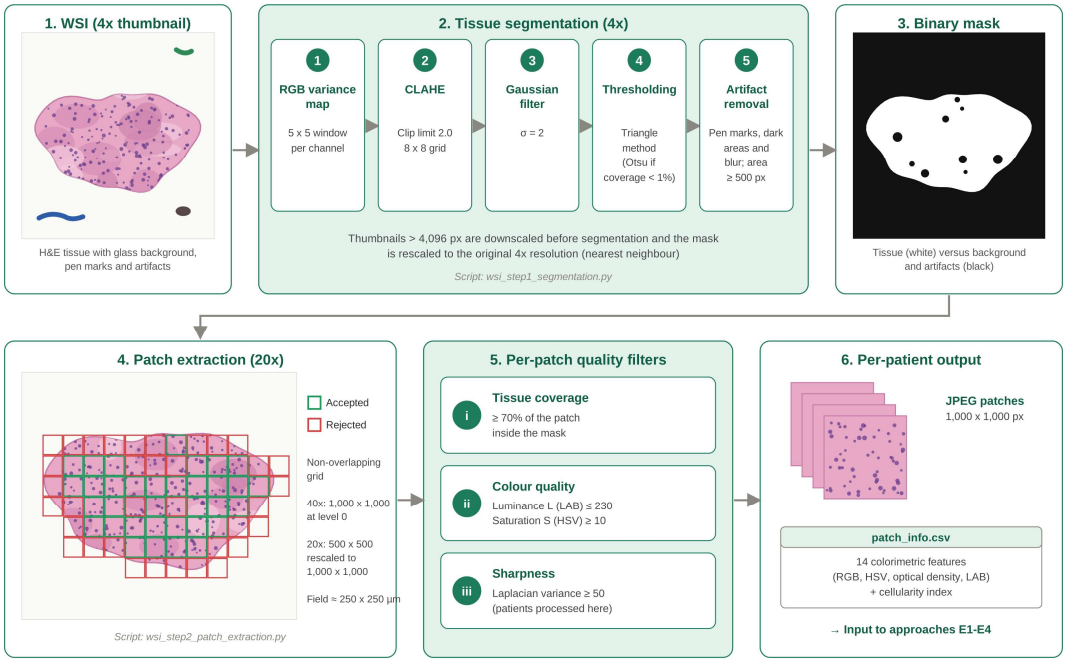


Figure S1. Common WSI preprocessing stages (Approaches 1 to 4): tissue segmentation at 4x in five steps, binary mask generation, extraction of 1,000 x 1,000 pixel patches at 20x without overlap, per-patch quality filters and per-patient output.

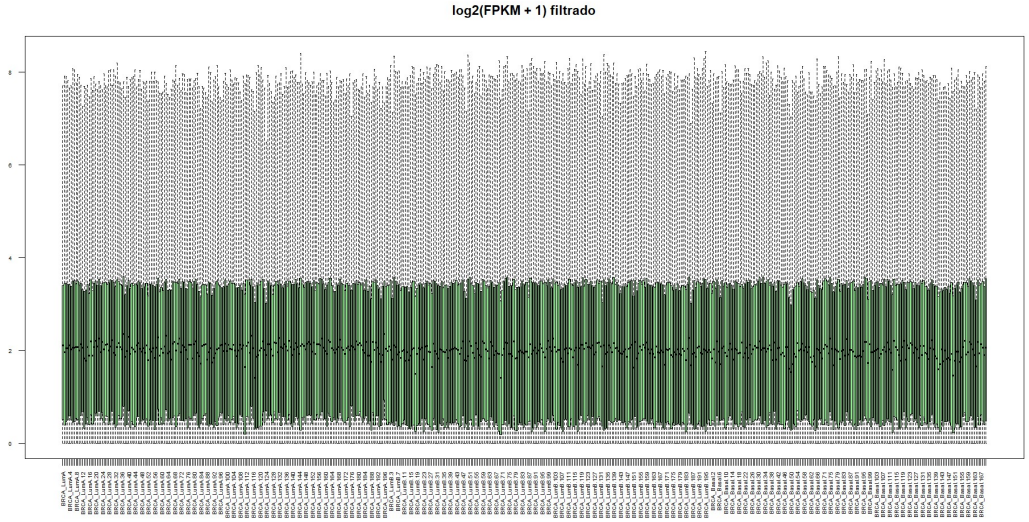


Figure S2. Distribution of gene expression per sample after the log2(FPKM + 1) transformation and gene filtering (TCGA-BRCA, n = 552). The homogeneous shape across samples confirms adequate normalization and comparability.

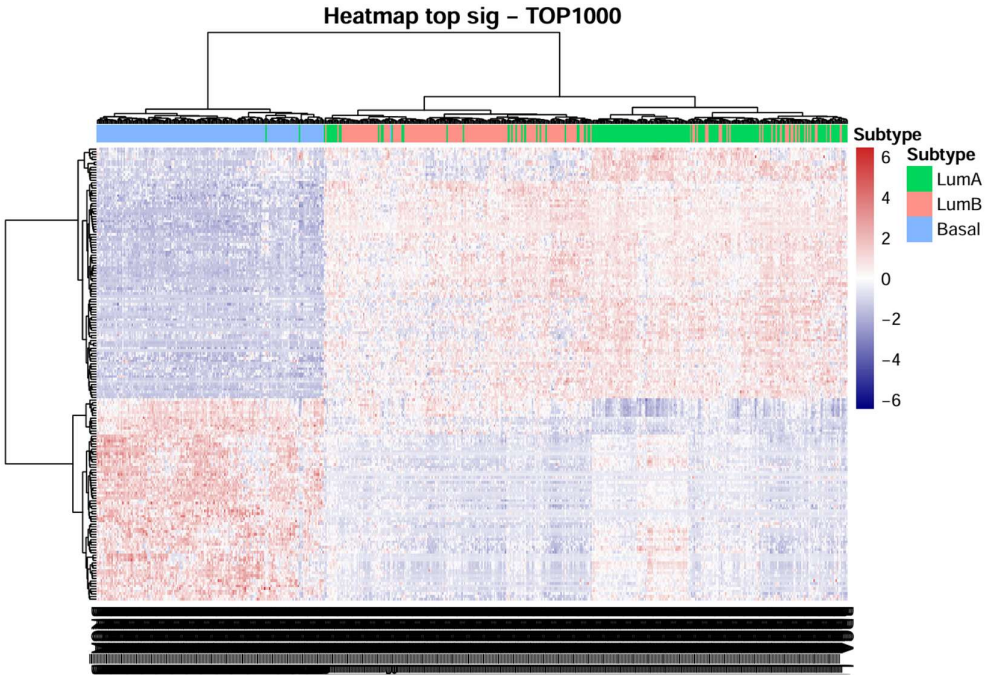


Figure S3. Heatmap of the differentially expressed genes in the one-vs-rest scheme (TOP1000), with hierarchical clustering of samples and genes.

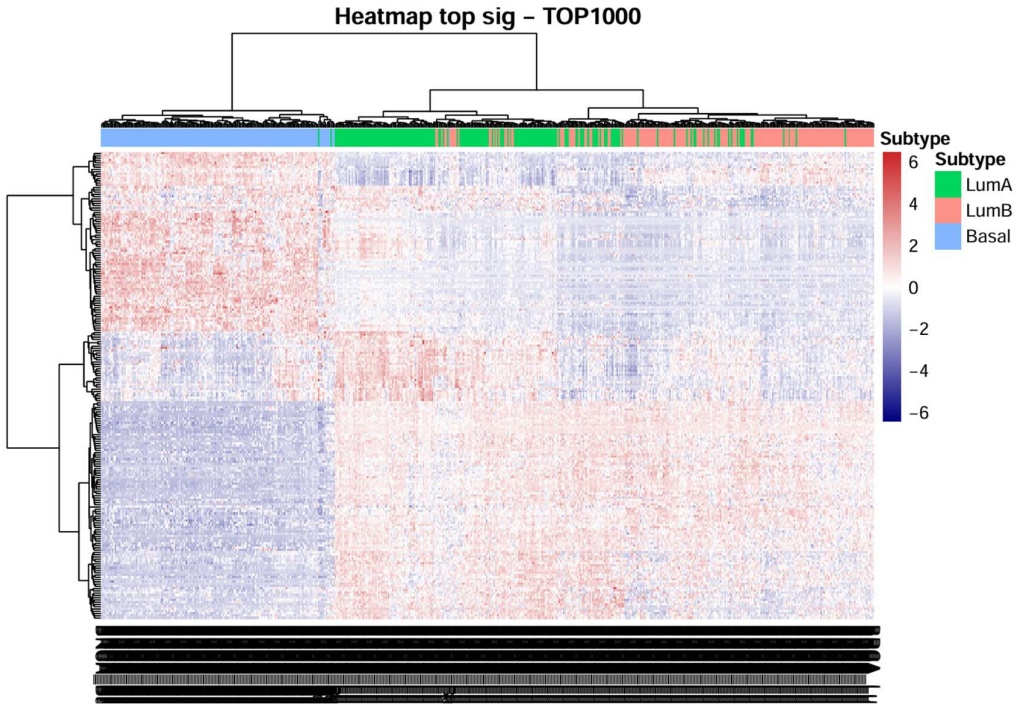


Figure S4. Heatmap of the differentially expressed genes in the one-vs-one scheme (TOP1000).

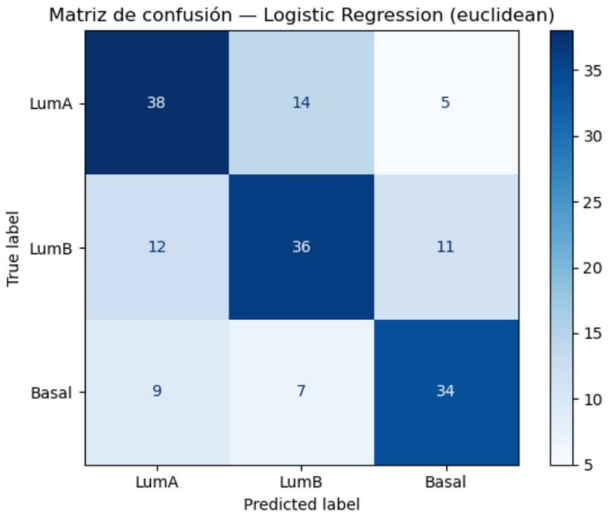


Figure S5. Confusion matrix and per-class metrics of Logistic Regression with the Euclidean metric (Approach E1) on the test set (n = 166): Luminal A, 38 of 57 correct; Luminal B, 36 of 59; Basal-like, 34 of 50.

	precision	recall	f1-score	support
LumA	0.64	0.67	0.66	57
LumB	0.63	0.61	0.62	59
Basal	0.68	0.68	0.68	50
accuracy			0.65	166
macro avg	0.65	0.65	0.65	166
weighted avg	0.65	0.65	0.65	166

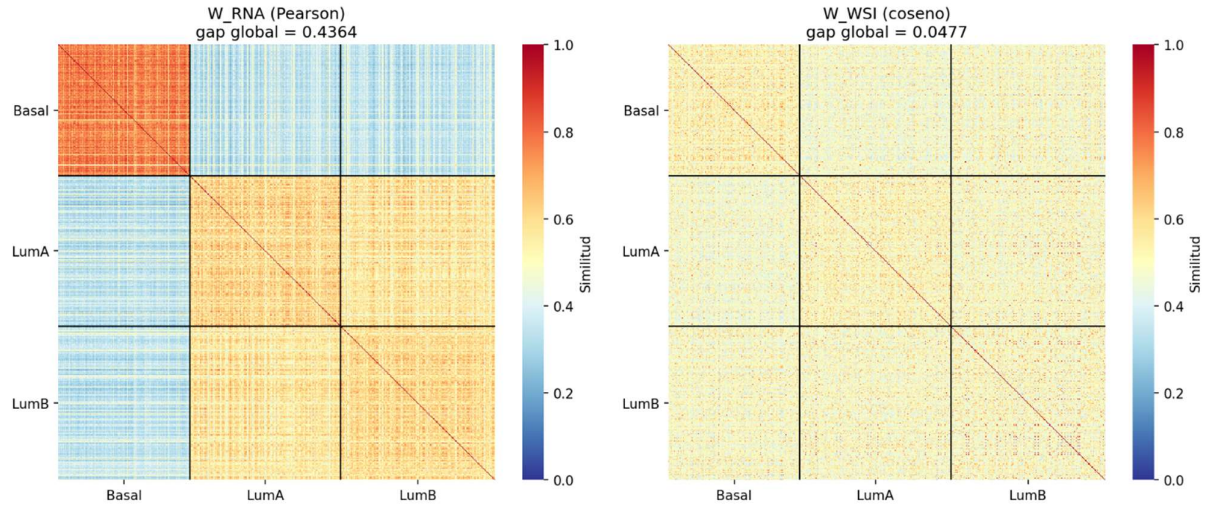


Figure S6. Patient-to-patient similarity matrices ordered by subtype. The transcriptomic matrix (left) shows three clear diagonal blocks (global gap = 0.4364); the histopathological matrix (right) shows a much weaker subtype structure (global gap = 0.0477). For display, similarities were rescaled to [0, 1] with W = (S + 1)/2; the gaps were computed on S.

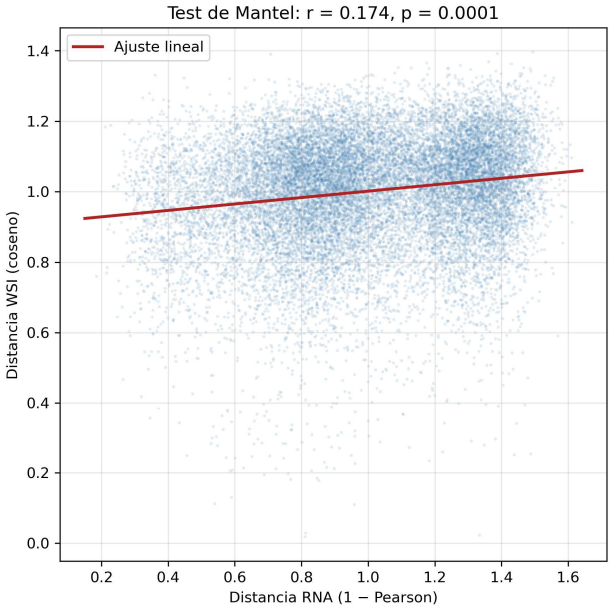


Figure S7. Agreement between the transcriptomic and histopathological (UNI-Crop, centered) distances of the 152,076 patient pairs, with the regression line. Mantel r = 0.174, p <= 0.0001. For legibility a random subsample of 20,000 pairs is plotted; the test uses all 152,076.

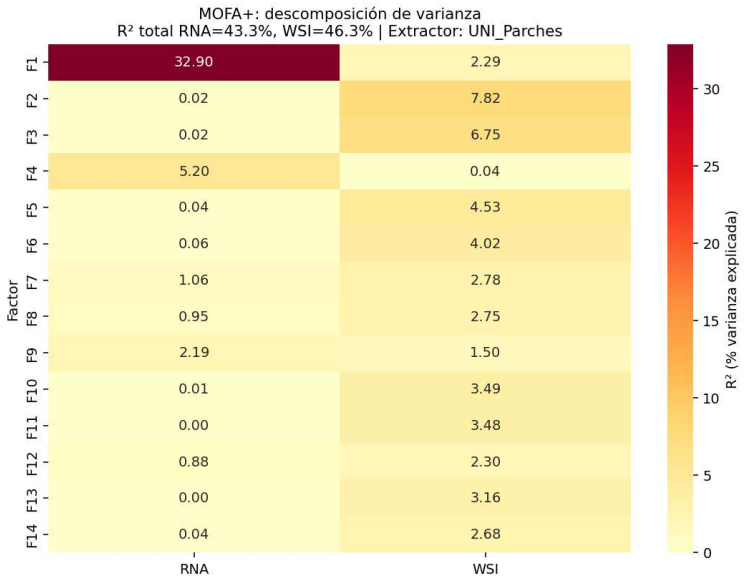


Figure S8. Variance explained (R2) by each MOFA+ factor in each view, as returned by MOFA+. The model explains 43.3% of the transcriptomic variance and 46.3% of the histopathological variance. Factor F1 concentrates the transcriptomic signal (32.90%), while the histopathological variance is spread across many factors, none above 7.82%. Eight of the fourteen factors were significantly associated with PAM50 subtype (Table S14).

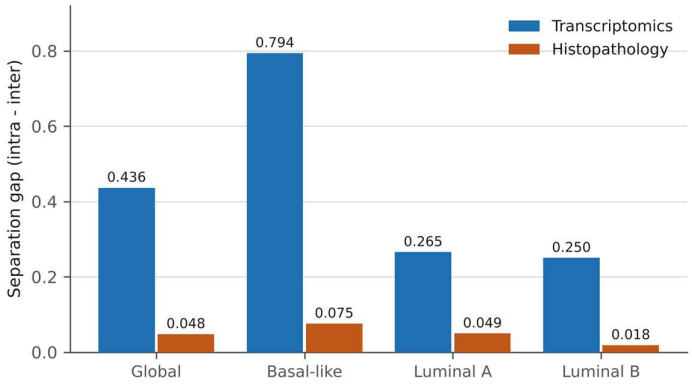


Figure S9. Subtype separation gap (mean intra-subtype minus mean inter-subtype similarity) in the two patient-to-patient similarity matrices. The global gap is the mean of the three subtype gaps. All comparisons were significant (Mann-Whitney U, $p < 0.001$).

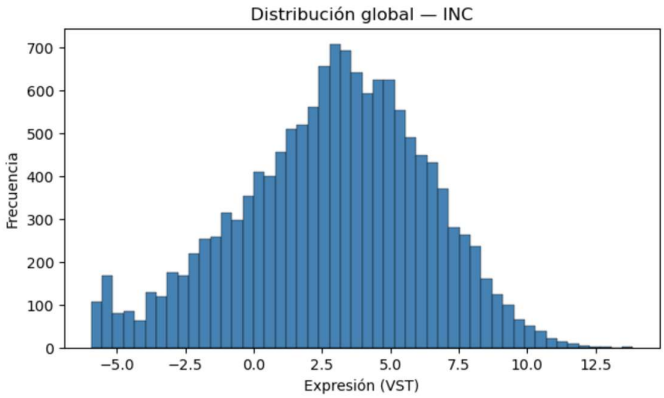


Figure S10. Global distribution of expression (VST) in the INC Bogota cohort.

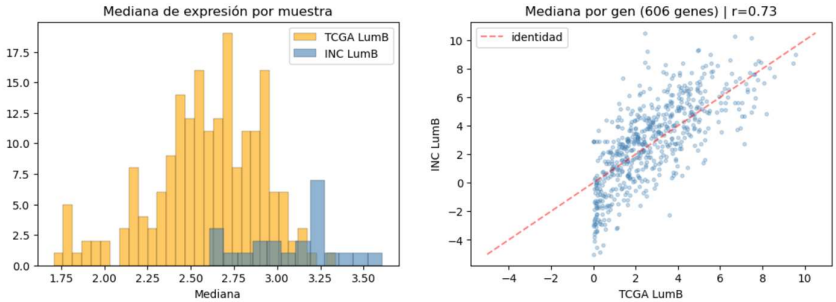


Figure S11. Comparison between TCGA-BRCA (Luminal B) and INC: median expression per sample (left) and median expression per gene (right), over the 595 signature genes present in both cohorts.

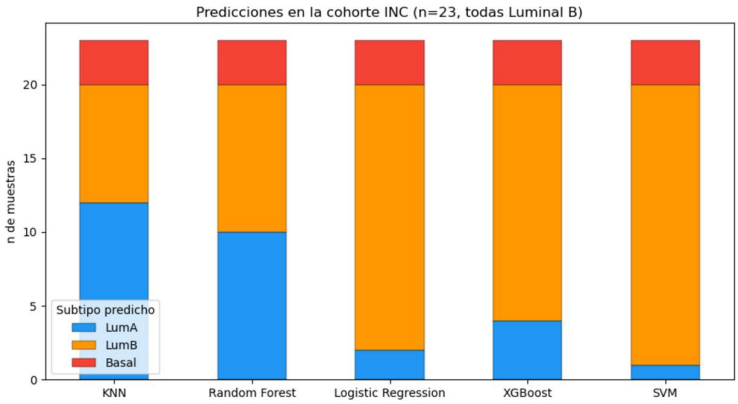


Figure S12. Distribution of the transcriptomic branch predictions in the INC cohort (n = 23, all Luminal B).